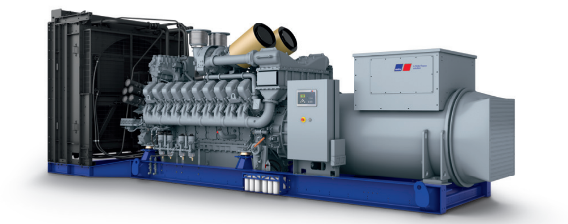
Diesel Generator Set

mtu 20V4000 DS3600

400 V - 11 kV/50 Hz/standby power/fuel consumption optimized 20V4000G94F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
— Low fuel consumption
— Optimized system integration ability
— High reliability
— High availability of power
— Long maintenance intervals

Support
— Global product support offered

Standards
— Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
— Generator set complies to ISO 8528
— Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
— NFPA 110

Power rating
— System ratings: 3580 kVA - 3730 kVA
— Accepts rated load in one step per NFPA 110*
— Generator set complies to G3 according to ISO 8528-5
— Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
— Engine-generator set tested to ISO 8528-5 for transient response
— 85% load factor
— Verified product design, quality and performance integrity
— All engine systems are prototype and factory tested

Complete range of accessories available
— Control panel
— Power panel
— Fuel system
— Fuel connections with shut-off valve mounted to base frame
— Starting/charging system
— Exhaust system
— Electrical driven radiators
— Medium and oversized voltage alternators
— Low voltage alternator

Emissions
— Fuel consumption optimized

Certifications
— CE certification option
— Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data[1)]

Engine		
Manufacturer		*mtu*
Model		20V4000G94F
Type		4-cycle
Arrangement		20V
Displacement: l		95.4
Bore: mm		170
Stroke: mm		210
Compression ratio		16.4
Rated speed: rpm		1500
Engine governor		ECU 9
Max power: kWm		3088
Air cleaner		dry

Fuel system	
Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption[2)]	l/hr	g/kwh
At 100% of power rating:	730	196
At 75% of power rating:	531	190
At 50% of power rating:	378	203

Liquid capacity (lubrication)	
Total oil system capacity: l	390
Engine jacket water capacity: l	260
Intercooler coolant capacity: l	50

Combustion air requirements	
Combustion air volume: m^3/s	4.3
Max. air intake restriction: mbar	30

Cooling/radiator system	
Coolant flow rate (HT circuit): m^3/hr	80
Coolant flow rate (LT circuit): m^3/hr	44
Heat rejection to coolant: kW	1090
Heat radiated to charge air cooling: kW	795
Heat radiated to ambient: kW	105
Fan power for electr. radiator (40°C): kW	105

Exhaust system	
Exhaust gas temp. (after engine, max.): °C	550
Exhaust gas temp. (before turbocharger): °C	643
Exhaust gas volume: m^3/s	10.6
Maximum allowable back pressure: mbar	50
Minimum allowable back pressure: mbar	–

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	Fuel consumption optimized without radiator		
		kWel	kVA*	AMPS
Leroy Somer LSA54.2 ZL17 (LV Leroy Somer standard)	400 V	2960	3700	5340
Leroy Somer LSA54.2 XL11 (Med. volt. Leroy Somer)	11 kV	2864	3580	188
Marathon 1040FDH7103 (Medium volt. marathon)	11 kV	2976	3720	195
Leroy Somer LSA54.2 ZL12 (MV Leroy Somer oversized)	11 kV	2864	3580	188
Marathon 1040FDH7105 (MV marathon oversized)	11 kV	2976	3720	195
Leroy Somer LSA54.2 ZL12 (Engine output optimized)	11 kV	2984	3730	195

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.